

# AimGel

## Artificial Cells for Immune Cell Activation

Animal free

Biomimetic

Chemically defined

T cells & NK cells

Human, murine & canine



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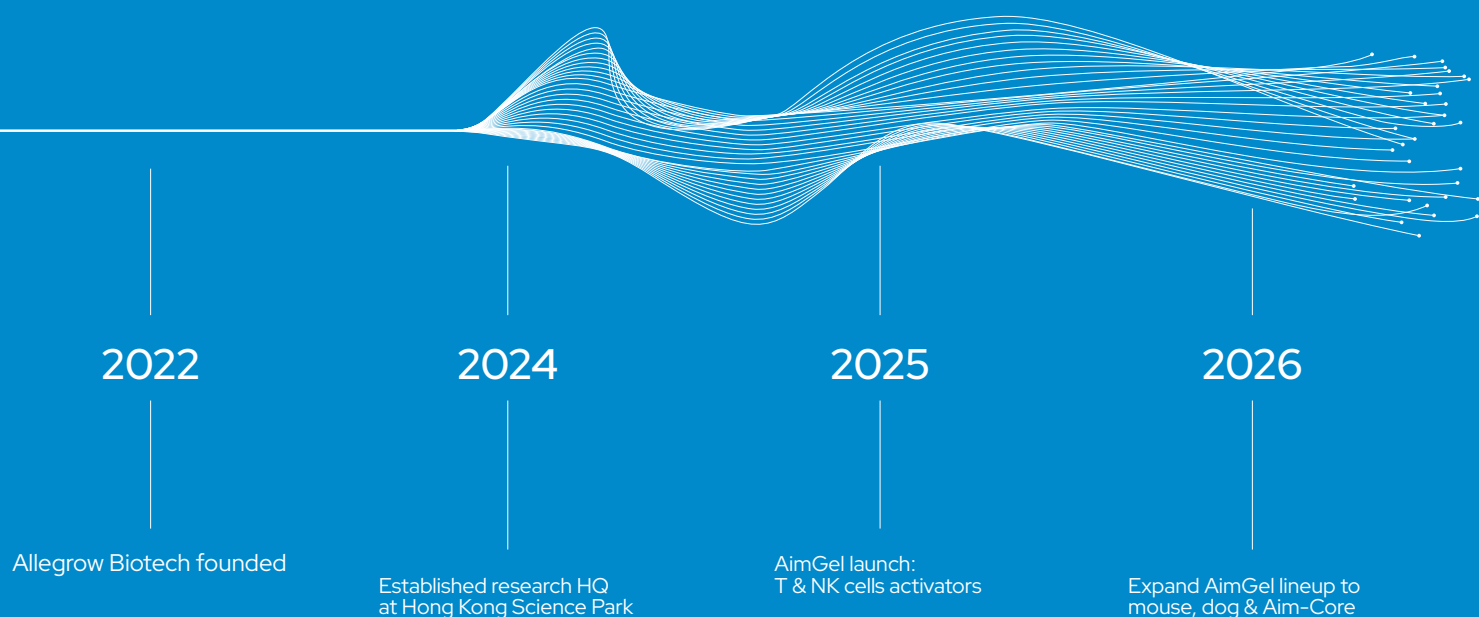


## About Us

Allegrow Biotech is a biotech specialising in immunomodulation via biomimetic approaches, based in Hong Kong Science Park, and spun out from the Hong Kong University of Science and Technology with their investment. Our team consists of experts in biomaterials, immunology and bioengineering, with more than 85% holding postgraduate degrees. We endeavour to open a new chapter in cell signalling and modulation with our AimGel technology.

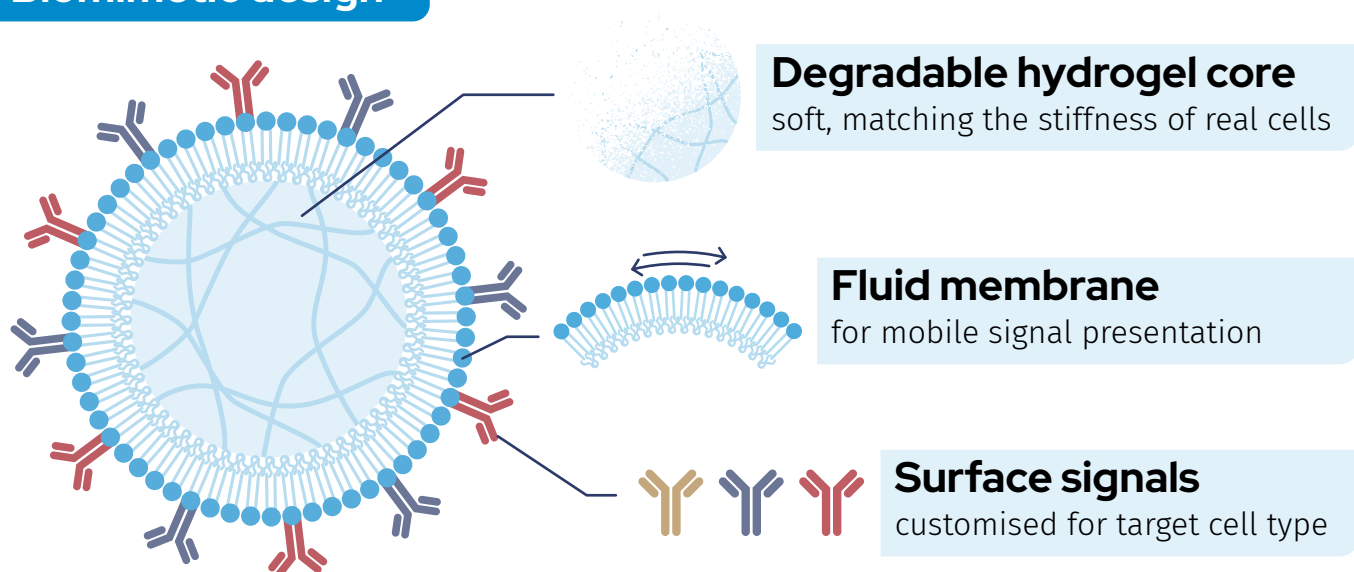
Our journey so far,

- ◆ Incubatee of Hong Kong Science and Technology Parks incubator programs - 2022 Ideation Program, 2024-2028 Incu-Bio Program
- ◆ Recipient of Research Talent Hub (RTH) and Technology Startup Support Scheme for Universities (TSSSU & TSSSU+) support
- ◆ Cohort 8, P4 Precision Medicine Accelerator
- ◆ Winner, HKUST-Sino One Million Dollar Entrepreneurship Competition 2023
- ◆ Winner, JumpStarter Global Pitch Competition 2023
- ◆ Gold Medal, 48th Geneva International Exhibition of Inventions 2023
- ◆ Gold Medal, Silicon Valley International Invention Festival 2024



# AimGel™ A modular artificial cell platform

## Biomimetic design



AimGel™ artificial cells are designed from the ground up to mimic natural antigen presenting cells (APCs). The soft hydrogel core replicates the mechanical stiffness and size (7-12  $\mu\text{m}$ ) of real cells, while activation signals move freely across the surface, just as they would on a natural APC membrane.

This biomimetic design delivers measurable advantages: AimGel™ requires significantly fewer activation signals than rigid magnetic microplastic beads while achieving more effective yet gentler activation. The result is higher cell expansion, better viability, and reduced exhaustion, cells remain responsive and can be re-stimulated.

## One platform, unlimited applications

Degradation kinetics, microparticle size, and signal loading capacity are all chemically tuneable, giving you precise control over activation parameters. Load different signals for different experimental needs, currently optimised for T cell (human, murine, canine) and NK cell (human) activation, with other cell types (Treg) and applications (subtype biasing) coming soon.

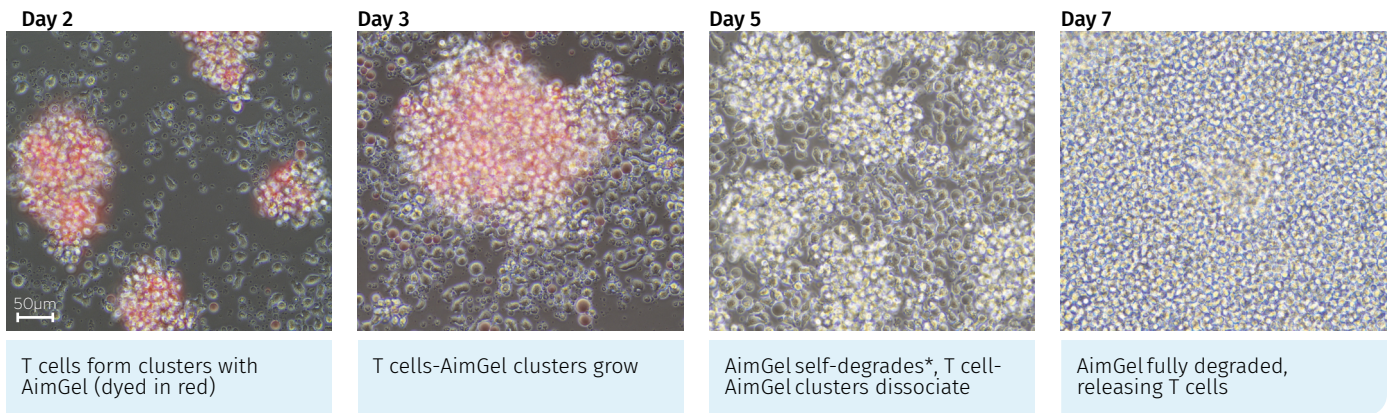
The platform adapts to your research: adjust activation duration, coordinate with lentiviral transduction timing, or engineer entirely new signal combinations using our Aim-Core lineup of blank microparticles.

Our AimGel activators ship non-hydrolysable and are cleared with the included lysis buffer. If your workflow needs them to clear on their own at a set point, we can formulate them to hydrolyse under your culture conditions.



# AimGel A modular artificial cell platform

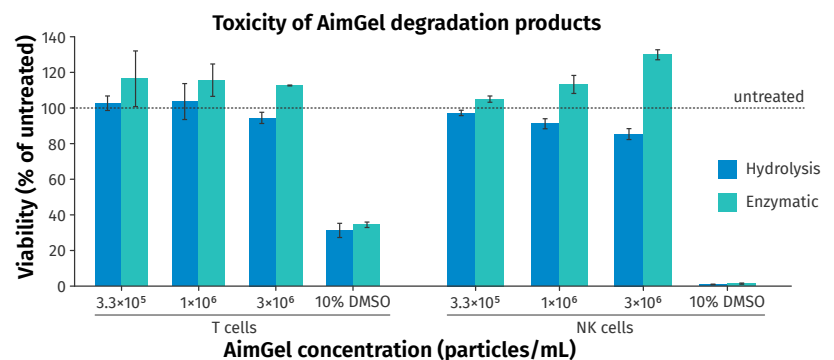
## Controlled, safe degradation



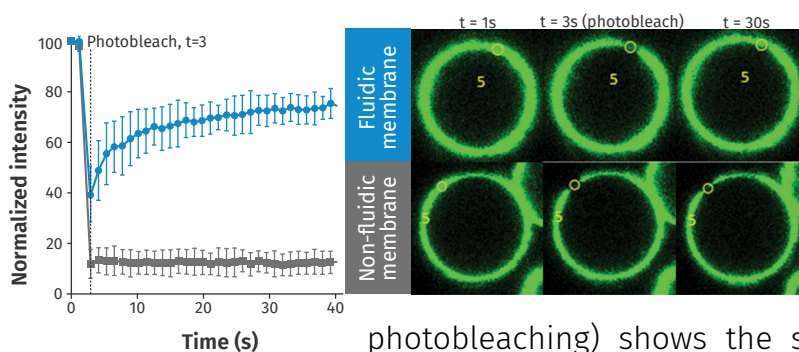
\*Shown with hydrolysable AimGel; our activators ship non-hydrolysable.

Removal takes one step. Add the included lysis buffer and the microparticles are fully degraded in 30 minutes at 37 °C. Dextranase clears the hydrogel core without affecting the protein signals or your cells. Nothing is separated out. The microparticles are digested in place rather than pulled away from your culture. No magnet, no column, and no cells lost to a separation step.

The degradation products from AimGel™ microparticles, be it from their own hydrolysis or the enzymatic degradation via the lysis buffer, do not affect human T & NK cell viability, compared to 10% DMSO cytotoxic control.



## Fluid, mobile membrane



On a rigid bead, signals stay where they are attached. On AimGel™ they diffuse freely, so surface ligands reorganise and localise as a cell engages, the way receptors form clusters at a natural immune synapse.

FRAP (fluorescence recovery after photobleaching) shows the signal recovery on a fluid membrane within seconds, while the non-fluidic control stays dark.

## Defined, consistent, compliant

AimGel™ is 100% synthetically derived with fully defined structures and ratios. No animal components means no batch-to-batch variability — just consistent, reproducible activation you can rely on for every experiment.

# AimGel™ A modular artificial cell platform

## Less prep



Mix & grow with one step enzymatic (dextranase) removal

### Magnetic beads

Magnetic separation, with potential cell loss

### Coated plates

Overnight coating then washing

### Feeder cells

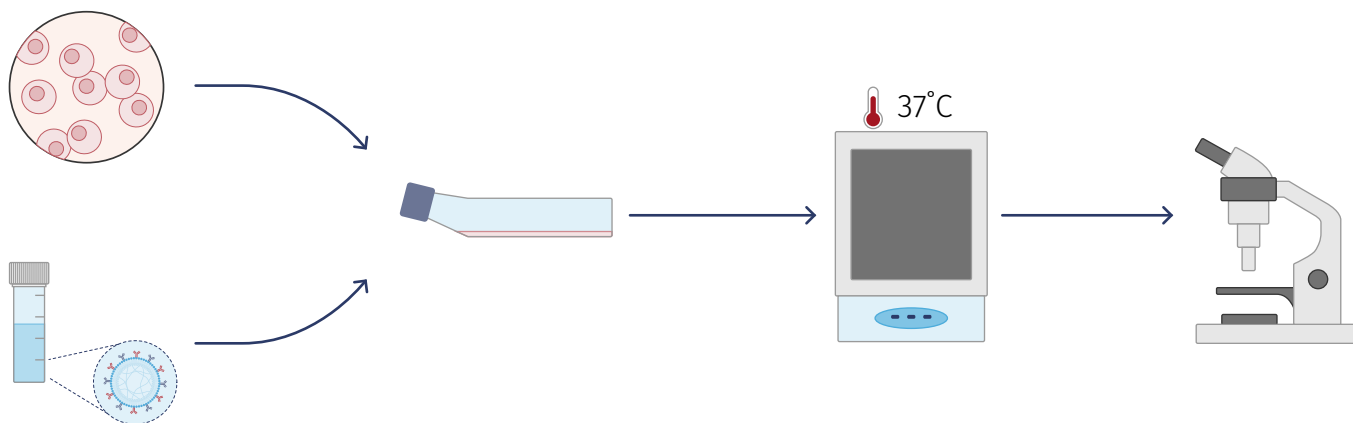
Cell-line maintenance  
irradiation/chemical cell killing

## Skip isolation steps

Whether you're starting with PBMCs or isolated cells<sup>1</sup>, AimGel™ activates both efficiently, achieving >90% activation in your target cell population.

1: Skips isolation steps from PBMC for T cells: human, mouse & dog. Isolation steps still recommended for NK cells.

## Sample workflow



- Step 1 Suspend AimGel™ with cells in culture vessel according to ratio
- Step 2 Incubate in your incubator of choice
- Observe Cells enter log growth phase the next day

You could also re-stimulate the cells according to your needs

# AimGel Human T Cell Activator

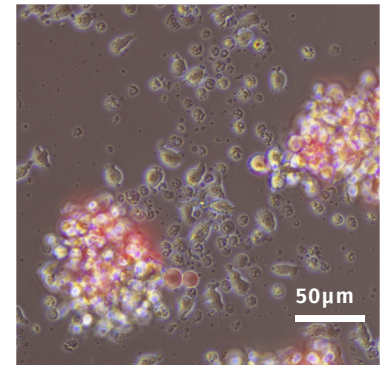
ATC01-0050H

## 3 signals

- anti-CD3
- anti-CD28
- anti-CD137

## Compatible with

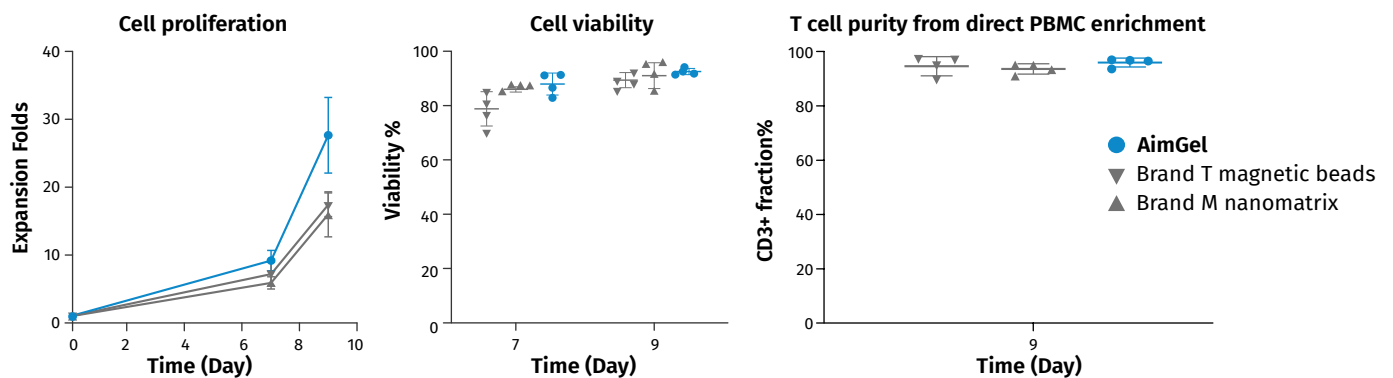
PBMC or isolated CD3+ T cells



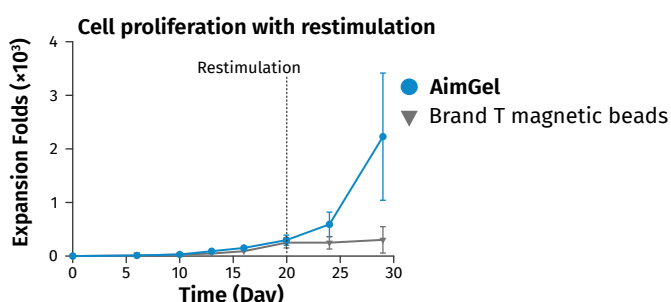
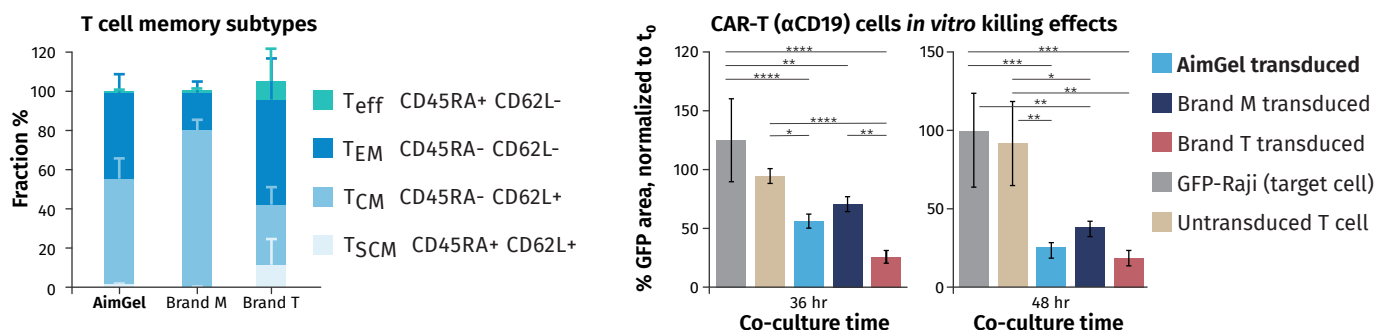
## Kit contents

1 vial 0.5mL at  $4 \times 10^7$  microparticles/mL ( $2 \times 10^7$  total), in PBS with 0.5% P/S  
1 vial lysis buffer: 100× dextranase in PBS

Activate and expand human T cells directly from PBMC. Get 60% more cells when expanding straight from PBMC when compared with magnetic beads and nanomatrix activators, with matching viability and purity.



AimGel produces more even fractions of effector memory and central memory subtypes. T cells expanded and activated by AimGel and then transduced showed matching cytotoxic effects compared with T cells expanded and activated by magnetic beads and nanomatrix.



With a gentler, more biomimetic stimulation that requires fewer signals/ligands, AimGel is able to restimulate cells quicker when compared to magnetic beads.

# AimGel Mouse T Cell Activator

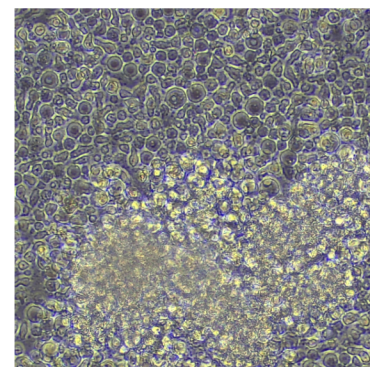
ATC01-0050M

## 2 signals

- mouse anti-CD3
- mouse anti-CD28

## Compatible with

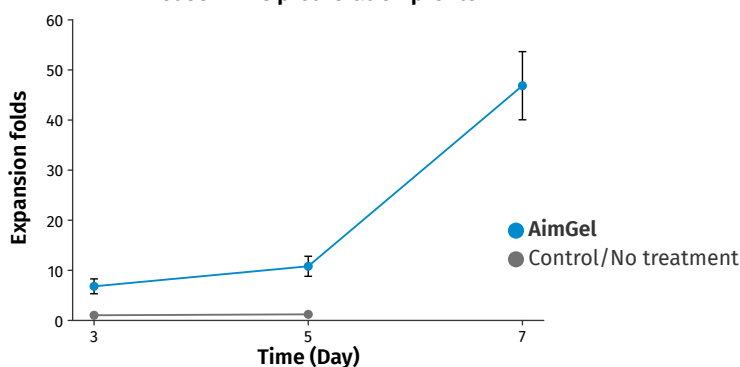
PBMC or isolated CD3+ mouse T cells



## Kit contents

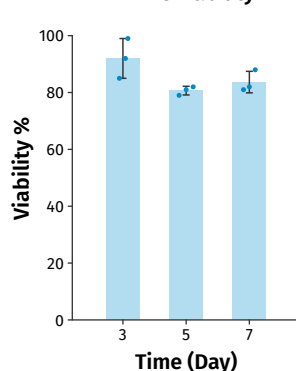
1 vial 0.5mL at  $4 \times 10^7$  microparticles/mL ( $2 \times 10^7$  total), in PBS with 0.5% P/S  
1 vial lysis buffer: 100× dextranase in PBS

## Mouse PBMC proliferation profile

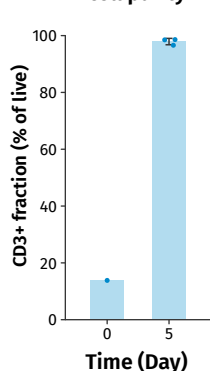


Activate and expand mouse T cells directly from PBMC, with no isolation step. AimGel™ treated PBMC reaches ~45-fold expansion by day 7.

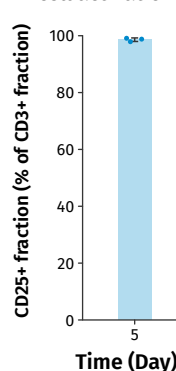
## PBMC viability



## T cell purity

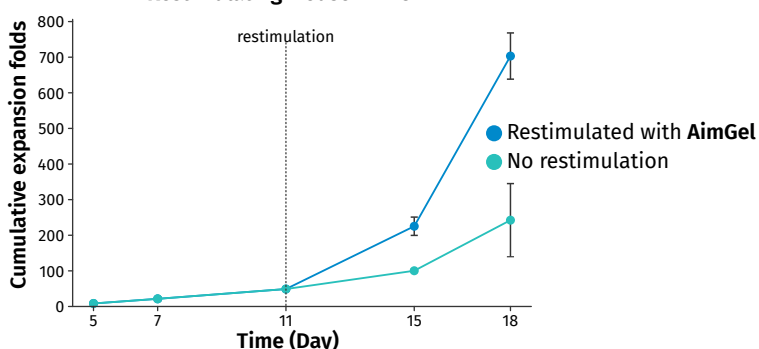


## T cell activation



AimGel also enriches: the CD3+ fraction rises from 14% to 98% by day 5, with 98% of those cells CD25+ activated. Factoring in PBMC expansion fold, T cell fraction expanded ~330-fold (approximating day 5 fraction for day 7).

## Restimulating Mouse PBMC



Re-stimulating at day 11 roughly triples the yield. Cumulative expansion reaches ~700-fold by day 18, against ~240-fold without re-stimulation. All cells were first activated with AimGel prior to re-stimulation.

# AimGel Dog T Cell Activator

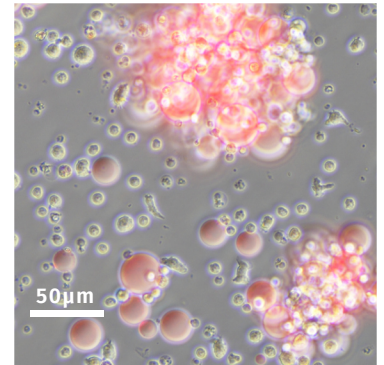
## 2 signals

- dog anti-CD3
- dog anti-CD28

## Compatible with

PBMC or isolated CD3+  
dog T cells

ATC01-0050D

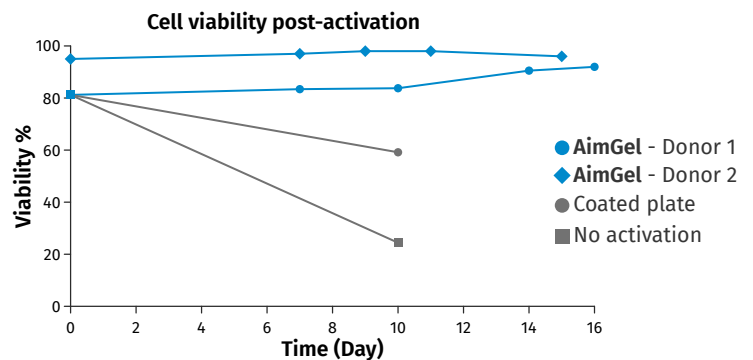
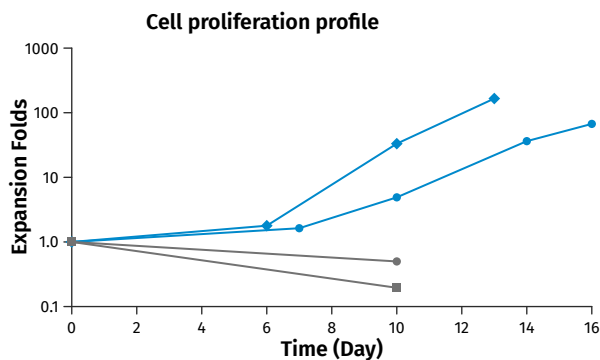


## Kit contents

1 vial 0.5mL at  $4 \times 10^7$  microparticles/mL ( $2 \times 10^7$  total), in PBS with 0.5% P/S  
1 vial lysis buffer: 100× dextranase in PBS

Expansion of dog PBMC with AimGel Dog T Cell Activator compared with well plates coated with dog anti-CD3/28 antibodies and negative control. By day 10, both coated-plate and unactivated cultures had fallen below their starting number. AimGel with dog anti-CD3/CD28 antibodies expanded 68-fold and 165-fold in two donors, by day 16 and day 13 respectively. Controls (coated plate and negative) done with donor 1 only.

Cells (PBMC) maintain high viability post-activation, with 2 dog donors. Compared with coated plate and no activation controls (donor 1 only).



# AimGel Human NK Cell Activator

ANK01-0050H

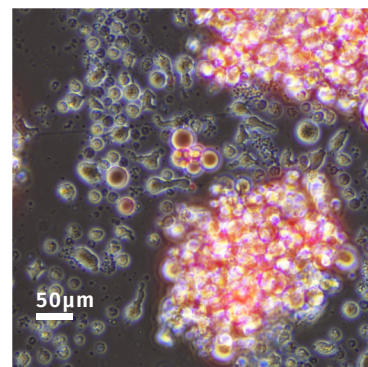
## Compatible with

Isolated human CD3- CD56+ NK cells

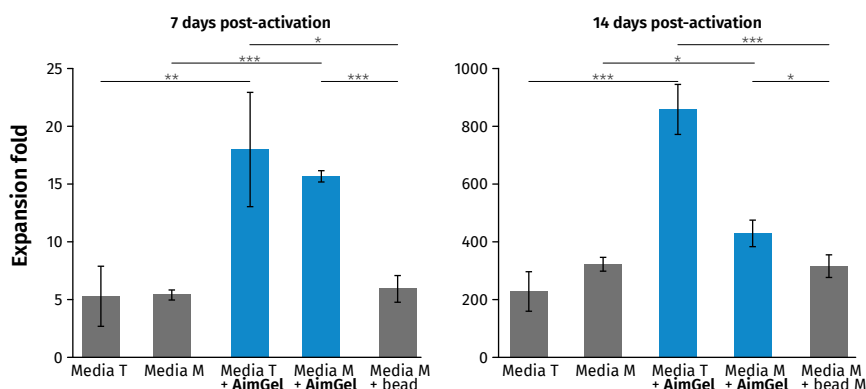
While culturing from PBMC is possible, it is dependent on the donor's starting NK fraction size.

## Kit contents

- 1 vial 0.5mL at  $4 \times 10^7$  microparticles/mL ( $2 \times 10^7$  total), in PBS with 0.5% P/S
- 1 vial lysis buffer: 100× dextranase in PBS

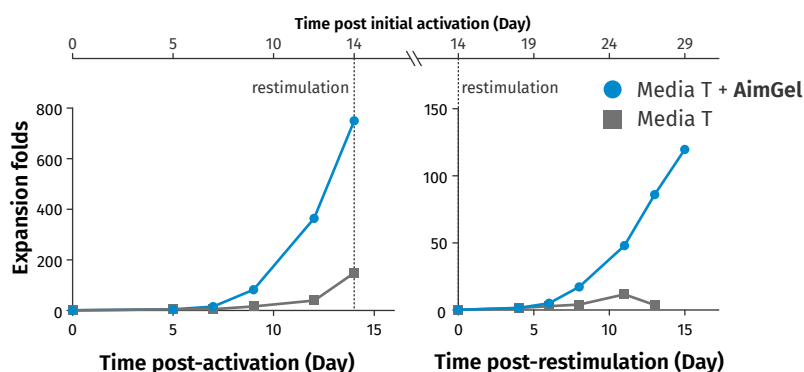


## NK cell expansion



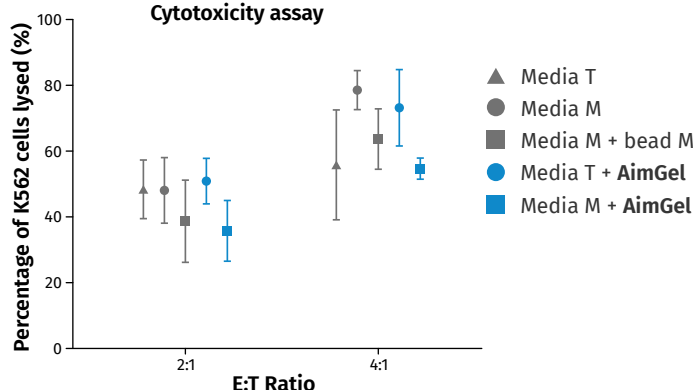
AimGel for human NK cells works with the NK media you already use. Added to two commercial NK media, it raised day-14 expansion from 230- to 860-fold in Media T, and from 325- to 430-fold in Media M. It also outperformed the bead activator sold alongside one of those media (M). Mean  $\pm$  SD, n=4. \* p<0.05, \*\* p<0.01, \*\*\* p<0.001.

## NK expansion over initial activation and restimulation



Re-stimulating at day 14 opens a second expansion phase. AimGel-activated NK cells expanded a further 120-fold over the following two weeks, where cells in medium alone peaked around 12-fold and then declined.

## Cytotoxicity assay



Cytotoxic function is unaffected. Against K562 targets at 2:1 and 4:1 effector-to-target ratios, AimGel-expanded NK cells lysed as effectively as cells grown in medium alone or with the bead kit. Function is preserved while yielding more cells.

# AimCore Blank artificial cells

AVIO1 – AVI12

## Signals

any biotinylated ligand

## Compatible with

Cell types and species of your choice

## Kit contents

1 vial 1.0 mL at  $4 \times 10^7$  microparticles/mL ( $4 \times 10^7$  total), in PBS with 0.5% P/S  
1 vial lysis buffer: 100× dextranase in PBS

Aim-Core is a blank canvas for cell signalling, enabling you to dock any biotinylated signals onto the surface for cell-like presentation.

Biotinylated signals just attach onto Aim-Core's fluid surface membrane via simple mixing.

We offer a selection of Aim-Core microparticles to give users choices in persistence in the culture media (degradability), signal loading capacity (avidin amount) and membrane fluidity.

## Degradability

transient (7-10 days) or persistent (&gt;60 days)

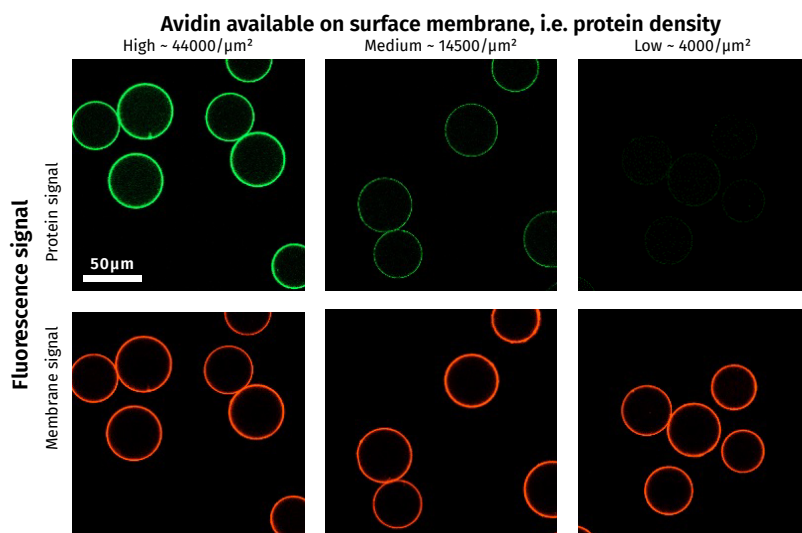
## Membrane fluidity

low or high

## Avidin loading

~4,000 / ~14,500 / ~44,000 per  $\mu\text{m}^2$ 

For example, our AimGel Dog T Cell Activator is based on the AVI11-0100 Aim-Core, which is persistent (non-hydrolysable), with high membrane fluidity and a medium signal loading capacity (~14,500 molecules of avidin per  $\mu\text{m}^2$ ). See our catalogue for the whole selection.



# Catalogue

## AimGel

|                         | Volume | Particles       | Cat. No.    |
|-------------------------|--------|-----------------|-------------|
| Human T Cell Activator  | 0.5 mL | $2 \times 10^7$ | ATC01-0050H |
| Mouse T Cell Activator  | 0.5 mL | $2 \times 10^7$ | ATC01-0050M |
| Dog T Cell Activator    | 0.5 mL | $2 \times 10^7$ | ATC01-0050D |
| Human NK Cell Activator | 0.5 mL | $2 \times 10^7$ | ANK01-0050H |

## AimCore

| Degradability | Membrane fluidity | Avidin amount               | Volume | Particles       | Cat. No.   |
|---------------|-------------------|-----------------------------|--------|-----------------|------------|
| Transient     | Low               | $\sim 4,000/\mu\text{m}^2$  | 1.0 mL | $4 \times 10^7$ | AVI01-0100 |
|               |                   | $\sim 14,500/\mu\text{m}^2$ | 1.0 mL | $4 \times 10^7$ | AVI02-0100 |
|               |                   | $\sim 44,000/\mu\text{m}^2$ | 1.0 mL | $4 \times 10^7$ | AVI03-0100 |
|               | High              | $\sim 4,000/\mu\text{m}^2$  | 1.0 mL | $4 \times 10^7$ | AVI04-0100 |
|               |                   | $\sim 14,500/\mu\text{m}^2$ | 1.0 mL | $4 \times 10^7$ | AVI05-0100 |
|               |                   | $\sim 44,000/\mu\text{m}^2$ | 1.0 mL | $4 \times 10^7$ | AVI06-0100 |
| Persistent    | Low               | $\sim 4,000/\mu\text{m}^2$  | 1.0 mL | $4 \times 10^7$ | AVI07-0100 |
|               |                   | $\sim 14,500/\mu\text{m}^2$ | 1.0 mL | $4 \times 10^7$ | AVI08-0100 |
|               |                   | $\sim 44,000/\mu\text{m}^2$ | 1.0 mL | $4 \times 10^7$ | AVI09-0100 |
|               | High              | $\sim 4,000/\mu\text{m}^2$  | 1.0 mL | $4 \times 10^7$ | AVI10-0100 |
|               |                   | $\sim 14,500/\mu\text{m}^2$ | 1.0 mL | $4 \times 10^7$ | AVI11-0100 |
|               |                   | $\sim 44,000/\mu\text{m}^2$ | 1.0 mL | $4 \times 10^7$ | AVI12-0100 |

## Not seeing your cell type or species?

Or want AimGel activators that self-hydrolyse on a schedule matched to your culture conditions? Reach out to [welovecells@allegrowbiotech.com](mailto:welovecells@allegrowbiotech.com) for custom formulations.

All items ship on ice and are stable at 4 °C for 12 months; contents sterile in unopened tube. Do not freeze. See individual protocols for handling limits.

See pricing at [www.allegrowbiotech.com/webstore](http://www.allegrowbiotech.com/webstore)

